



Kawal Banga

Moderator

27 Jan 2026 13:52

Helpful Hints

Hi,

This thread will contain helpful hints - hints to ease your journey through TM470.

Feel free to add any useful links or resources that you come across in your research that could help other students.

Regards

Kawal



Kawal Banga

Moderator

27 Jan 2026 14:00

Edited by the author on 27 Jan 2026 14:00



Making use of Microsoft Word

Many of my TM470 students have struggled to make effective use of some of the facilities in MS Word. So over the years I have produced some guidance documents. Some of these contain links to other resources. (Being able to produce good presentable documents is an employability skill. So worth improving MS Word skills and knowledge, if appropriate.)



Effective use of MS Word 25-03-19 - Forum Copy.zip



Gary Waddell

15 Feb 2026 10:39



Re: Making use of Microsoft Word

Hey @Kawal thanks for the useful documents. It's been a while since I've studied at the OU, are there any specific styles we should use/import when submitting for TMA/EMA?



Kawal Banga

Moderator

15 Feb 2026 22:01

Edited by the author on 15 Feb 2026 22:08



Re: Making use of Microsoft Word

Hi Gary,

There are no detailed templates or document styles for TMAs/EMAs on the TM470 module website. We have 4 specialisms: Software Engineering, Cyber, Networking and Data, and we can have 3 different styles of project: development, evaluation or research. That's 12 variations. And of course every project is unique.

There is a basic "TMA and EMA template" and also a resource "Writing, Structuring, Styling and Editing Reports" under resources.

If you need additional guidance, you should discuss this with your tutor.

Kawal



Kawal Banga

Moderator

27 Jan 2026 14:33

Edited by the author on 4 Feb 2026 11:40



Managing Word count

Two useful resources (with additional URL links) to help you manage your word count and write concisely and succinctly.

Edit: apologies. I can't count! I've added the second resource, the PowerPoint slides "Tight Essays (keeping within Wordcounts)", which are excellent.

(Being able to write concisely and succinctly is an employability skill. So worth improving this skill if appropriate.)



Tight Essays (keeping within Wordcounts).ppt



Word counts v1.4 - TM470.pdf



Alisdair P

2 Feb 2026 10:03



Re: Managing Word count

Hello Kawal,

Thank you for this resource. This will be helpful for me to determine I should go about each assignment. One question I have when you say text in headings is excluded from the word count, does this also include titles for each section of a TMA?



Kawal Banga

Moderator

2 Feb 2026 10:09



Re: Managing Word count

Hi Alisdair,

Yes. All titles/headings.

Kawal



Alisdair P

2 Feb 2026 10:10



Re: Managing Word count

Excellent, thank you very much.



Kawal Banga

Moderator

27 Jan 2026 14:35



Developing a Framework (process or model) -and evaluating it

Developing a Framework (process or model)

Some students develop a framework for something or other. Please remember, that you will still need stakeholders from which to gather requirements, and they will also need to evaluate your final framework. Developing a framework will still need to go through the cycle of requirements, design, build, test and implement. Unfortunately, the implementation phase is often missing from such student projects.

Evaluation and testing are not necessarily the same thing. A distinction: Testing is what happens during development, whereas evaluation is more focussed on assessing that what you set out to achieve/deliver has indeed been delivered, and that it works (is fit for purpose). It is the stakeholders/end users who will decide if it meets their needs through some evaluation process. You cannot really evaluate your own work (something many students try to do!).

Getting some stakeholders to pilot the framework in a real environment and getting feedback is one way of evaluating it. Doing a survey of users, or other experts, and getting their views on the framework might be another way of

evaluating it, albeit piloting the framework is usually a stronger evaluative process than simply asking for feedback from reading a document. Doing both (piloting & getting feedback) is preferred.

(UAT – user acceptance testing – by end users is a form of evaluation.)

The framework is typically documented as a Word document. Students often neglect to also include a document on how to implement the framework. This should really be an essential component (Implementation Guide) of the framework delivery. And guess what? You need to evaluate the implementation guide too!

(These comments also apply to those developing an IT/IS process or an IT/IS model.)

(These are my views, and others may not agree with me!)

Kawal Banga



Kawal Banga

Moderator

27 Jan 2026 14:37



Development projects - not just code!

Development projects should also deliver some combination of user documentation, maintenance guides, implementation guides. And these should also be tested/evaluated.



Kawal Banga

Moderator

2 Feb 2026 15:47



Including sample code snippets for those doing software projects

Hi

We just need some samples of code. Please **do not** include all of your code. You can do a copy and paste of code or show screenshots. So you might want to show some code which is complicated, does something tricky or you're proud of for whatever reason. But make sure you briefly explain what the code does and why it has been included.

You should list all of the source programs you have created in a table in an appendix along with a short description, and also include some kind of structure diagram in the main chapter showing the relationship between the source programs/modules.

Also, don't include code on a black background, as some tutors print off TMAs to mark them. Thanks.

Kawal



Kawal Banga

Moderator

27 Jan 2026 14:41

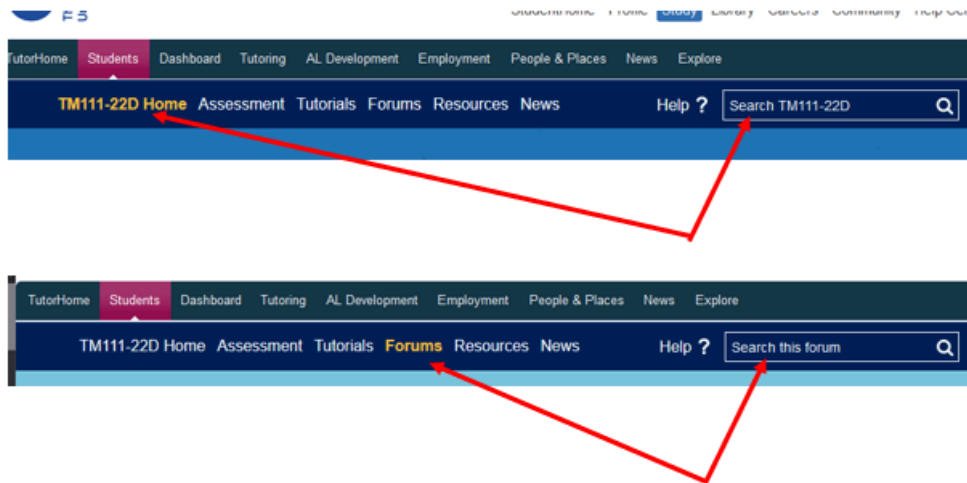


Using the search field on module websites and forums

Hi,

There is a search field in the top right hand corner on every screen on the module website including the forums. If you type "word count", TMA02, "question 2", etc, you will find what you are looking for if it exists somewhere.

Please search before you post anything, as your question may already have been addressed in an existing post. Thanks.



Kawal Banga

Moderator

27 Jan 2026 14:53



Harvard Referencing and Citations

Hi,

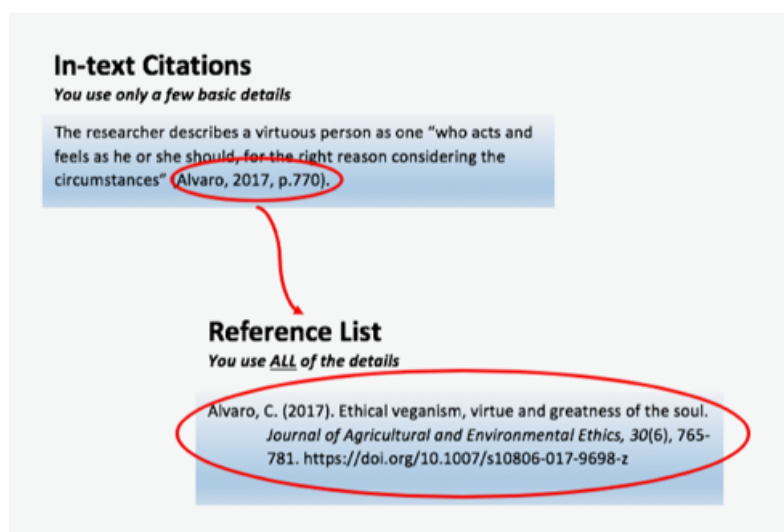
You should all be familiar with Harvard Referencing and Citations, this being your final module. Most of what you need is at the following link for the "Quick guide to Harvard referencing (Cite Them Right)".

<https://www5.open.ac.uk/library/referencing-and-plagiarism/quick-guide-to-harvard-referencing-cite-them-right?nocache=67c47985750fd>

Please make sure you have searched here before posting any query on referencing. Thanks for your understanding.

Reference v citation

A reference is listed in the references list at the end of your document (before any appendices) and a citation (where you cite your reference) is in the body of your text, sometimes referred to as in-text citation. You need both.



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Kawal Banga
Moderator
27 Jan 2026 14:55



Embedding documents in Word TMA/EMA documents - please don't – and only submit 1 file

Embedding documents in Word TMA/EMA documents

Never embed any document (Word, Excel, Powerpoint, PDF, etc) into your TMA/EMA Word document. There is no guarantee that the recipient will be able to open an embedded document. For Word documents, you can copy and paste the text into appendices - but might need a bit of re-formatting. Excel, PDF and PowerPoint will probably mean taking screenshots and putting them into appendices. You can paste Excel cells into a Word document, and it converts them into a Word Table by default - these often require reformatting.

Only submit a single file - a Word document

Also, remember that "Your TMA01 should be written in Microsoft Word and only one file may be submitted for assessment. Please do not submit files in other formats, or additional files containing code for example. Please use landscape format for some pages where it improves readability for the marker – this can be a problem for some screen shots and program code listings."

If in doubt, please ask your tutor.

SW

Sean Wright
21 Feb 2026 15:17



Re: Embedding documents in Word TMA/EMA documents - please don't – and only submit 1 file

Hi Kawal,

Just a quick question, if submitting one document and we want to add our consent forms and participation information sheets, would these be counted in the advisable word count? (I'm aware we aren't marked on word counts.)

Or would these be added in an appendix at the end and be classed as supporting documents?

Thanks



Ju Tope
Moderator
21 Feb 2026 17:08



Re: Embedding documents in Word TMA/EMA documents - please don't – and only submit 1 file

Consent forms etc go in appendices, these are additional evidence referenced from main report not supporting docs in the way I think you mean

SW

Sean Wright
23 Feb 2026 20:25



Re: Embedding documents in Word TMA/EMA documents - please don't – and only submit 1 file

Thanks for getting back to me Ju.

That definitely clears it up for me.



Kawal Banga
Moderator
27 Jan 2026 15:00



Creating links in Word Documents - and links to and from appendices

If you have a lot of appendices, then provide return links back from the appendices to where you referred the reader to the appendix from your main chapters. This improves navigation and readability.

Tutors are not obliged to read appendices, but only to scan them. Appendices are supposed to contain supporting

information. You need to make sure that you have a good descriptive narrative in your main chapters for any appendices you are referring the reader to. So don't just have a heading and say "See appendix C".

You can of course create links for any headings, and not just appendices.



Inserting links in MS Word documents v 1.1.pdf



Kawal Banga

Moderator

27 Jan 2026 15:12



Showing changes as you move from TMA01 to TMA02 and TMA02 to TMA03

Hi,

It would be a good idea to show what has been added, deleted or changed when going from TMA01 to TMA02 and TMA02 to TMA03. How you do that, you need to agree with your tutor. You could use different colours, or the MS Word revision feature. But please contact your tutor.

And remember that TMA02 builds on TMA01; TMA03. builds on TMA02; and your final EMA will build on your TMA03. So each assignment contains material that is reusable for the subsequent ones and the final one.

SH

Stephen Healey

13 Mar 2026 20:18



Re: Showing changes as you move from TMA01 to TMA02 and TMA02 to TMA03

Hi,

Thanks for the clarification. I just want to double-check that I've understood correctly.

For TMA02, should I take my existing TMA01 document and **continue developing it**, adding or removing sections as needed, rather than starting a completely new document?

Also, the word limit for TMA01 was 2,000 words and TMA02 has a 4,000-word limit.

Does that mean TMA02 should effectively build on top of TMA01 so that the total grows to around 4,000 words (instead of adding an extra 4,000 words on top of the original 2,000)?

Just want to confirm how the word count should be handled before I proceed.

Thanks,

Stephen



Kawal Banga

Moderator

14 Mar 2026 01:51



Re: Showing changes as you move from TMA01 to TMA02 and TMA02 to TMA03

For TMA02, should I take my existing TMA01 document and **continue developing it**, adding or removing sections as needed, **Yes, this is correct**

Does that mean TMA02 should effectively build on top of TMA01 so that the total grows to around 4,000 words **Yes, this is correct, but you might move some TMA01 content to appendices, and there might even be some content you delete from TMA01**

Remember, the word counts are guidance only.



Kawal Banga

Moderator

27 Jan 2026 15:26



TM470 projects 3 types: Development, Evaluation and Research projects.

I've prepared this short document primarily to help those students who are still thinking about project ideas.

TM470 projects can be categorised into three main types: Development, Evaluation and Research projects. I've extracted the text referring to these from the Project Choices document and the TMAs and EMA below to help you understand what might be required for each, but you should still read these documents in full. (I've colour coded them so you can follow the text for any one of them with ease.)

I've also attached a document to help students who are doing research type projects.

These are resources I have prepared for my own TM470 students, but I am sharing them here amongst all students. If you have any queries then please discuss them with your own tutor.



TM470 3 project types - extracts from TMAs EMA.pdf



TM470 Research Projects - v 1.1.pdf



Kawal Banga

Moderator

10 Feb 2026 11:23



Books on research available in OU library

Books on research available in OU library

There are research books available in the OU library. Here's 3 examples I found just now.

Polonsky, M. J. and Waller, D. S. 2019. Introduction. In: Designing and Managing a Research Project: A Business Student's Guide. Fourth Edition edn, Thousand Oaks, CA: SAGE Publications, Inc. pp. 3-17 Available at: <<https://doi.org/10.4135/9781544316499.n1>>; [Accessed 10 Feb 2026].

Sharp, JA, Peters, J, & Howard, K 2002, The Management of a Student Research Project, Taylor & Francis Group, Oxford. Available from: ProQuest Ebook Central. [10 February 2026].

Kuada, J 2012, Research Methodology, Samfundslitteratur, Frederiksberg. Available from: ProQuest Ebook Central. [10 February 2026].

Also, <https://methods-sagepub-com.libezproxy.open.ac.uk/books-and-reference> is a very useful resource for searching on any research topics.



Kawal Banga

Moderator

27 Jan 2026 15:50



Your project journey (according to Kawal Banga)

We're interested in your project journey as follows:

- 1 You identified a real business/social problem that could be solved through an ICT solution (LO2), engaging with sponsors/users who needed a solution to the problem.
2. You gained ethics approval for your project, and modified your project if appropriate to gain approval (LO10).
3. You project managed (keeping evidence of records, plans, outcomes) the delivery using a suitable project/process lifecycle (LO9).
4. You identified and managed risks (LO3) on the way and identified and utilised skills, knowledge, resources and people you needed (LO3).
5. You made use of technical concepts and principles (LO1) from your Level 3 modules.

6. You analysed, designed and developed an ICT solution building on and extending skills from your Level 3 and other modules (or equivalent professional skills), and using any additional skills you needed (LO10).
7. You took into consideration any LSEPIs (Legal, Social, Ethical, Professional issues) and EDI (Equality, Diversity and Inclusion) issues and modified your project and your behaviour to deal with such issues (LO10).
8. You carried out a literature review using quality, credible and relevant sources in which to ground your work, and supporting your decisions (LO4, LO6).
9. You worked independently as much as possible and learned new skills and knowledge that you applied to your project (LO8). You also understand how you learn and study effectively, as well as overcoming any challenges that lead to ineffective learning and study (LO8).
10. You reflected on things (processes, tools, resources, studying, etc) that worked or things that didn't work (LO5), and lessons and skills (technical, professional, academic, organisational, project management) that you learned through the project.
11. You replanned and rescheduled your work when things went wrong (LO9, LO3, LO5, LO8).
12. You communicated effectively through TMAs/EMA, reports, emails etc with your tutor and other project stakeholders (LO7).
13. You engaged the sponsors and/or users throughout the project journey, where appropriate, seeking feedback on interim deliverables and they evaluated your final artefact (LO7, LO11).
14. You can prove all of the above with solid evidence that you collected over the project journey, and can communicate this effectively to your tutor and other stakeholders (LO7).

All choices and decisions are yours, which you make through your previous knowledge on the L3 modules and any additional research you do. You need to be able to justify all such choices and decisions in your TMAs and EMA, and critically reflect on these (LO5, LO8).



Kawal Banga

Moderator

27 Jan 2026 16:05



Developing a solution to a real business/social problem v a "made up" project

Developing a solution to a real business/social problem will offer you the best opportunities to get higher grades against the 11 LO (Learning Outcomes). Also, developing a solution where one does not already exist.

An approach whereby you come up with a project (a made-up project!) that is technology skills driven, to demonstrate your technical skills, or just aimed at getting you a pass through the module, will not offer the best opportunities to get high grades against the LOs.

Despite my comments about such projects being 'made-up', it doesn't mean you won't pass the module. Every project is different, and every project student is different, so it's always difficult to generalise.

Some wiser words from a fellow tutor:

I also emphasise to my students that they should look for an approach that maximises their ability to demonstrate their skills and not to minimise everything to the point that it becomes a trivial exercise.

This trivialisation is very frequently apparent in the approach to legal, ethical, accessibility, security etc issues.

Investigating real business or social problems with real people in a real problem area they know about makes the best projects.

Fictional projects = [potentially] lower grades and [higher] possibility of failure.

Having a project sponsor

If you're running a project within and for an organisation you will usually have a project sponsor. As you know, the sponsor typically provides resources such as money and access to other people as well as guidance, support, steering and direction to the project via the project manager. The sponsor (representing the end-users) is the one pushing for the project and will chase the project manager (you!) to deliver. (NB This is very simplistic representation of what a sponsor does.)

Many student projects don't have a sponsor which often means that there is no one who really needs the artefact from the project. Sometimes there is no sponsor, but there are key end-users, or individuals, who do want the artefact.

Benefits of developing a solution to a real business/social problem

By doing a real project you have opportunities for learning and reflecting, as well as real-life project experiences based on things such as

- Sponsors and end users changing their minds
- Changing project dates
- Sponsors/resources going on holiday for two weeks so you can't contact him about urgent project issues.
- Technical support resources, or facilities, becoming unavailable and/or being given to higher priority projects.
- Changing their requirements (or recognising that that's not what they meant!)
- New requirements
- Highlighting missing requirements, the week before go live
- Changing their priorities
- Reducing the budget for the project
- Cancelling the project!

This gives you real-life experience of dealing with things such as the following:

- Change management
- Risk management
- Replanning
- Prioritising
- Re-scoping or de-scoping the project.
- Problems with resources (people and other resources).

This is all very good experience which you will get more from a real project than a made-up project, enabling you to reflect more and learn more which will improve your grades for your learning outcomes.



Kawal Banga

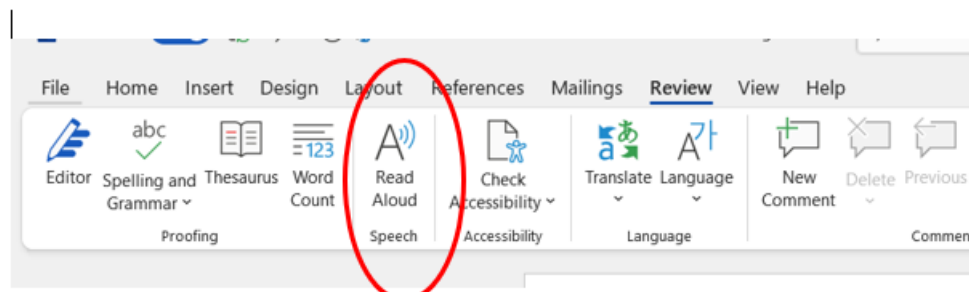
Moderator

4 Feb 2026 11:45



Microsoft Word "Read Aloud" to check the readability of your work

Use the Word "Read Aloud" facility to hear if what you have written reads well or not.



Kawal Banga

Moderator

4 Feb 2026 12:00



Using the Microsoft Voice Access (speech recognition) facility for dictating your work

I try to use this as often as I can for emails, producing any documents, and providing feedback to students' assignments. I find that I can dictate about 10 times faster than I can write by typing. It's not perfect, but even after making any changes (usually punctuation) it's still much faster.

For Windows 10, press the Microsoft logo key (Windows Key) and the H key simultaneously.



For Windows 11, go to Settings > Accessibility > Speech ...

Lots of resources online for this.

Get started with voice access

<https://support.microsoft.com/en-gb/topic/get-started-with-voice-access-bd2aa2dc-46c2-486c-93ae-3d75f7d053a4>



Kawal Banga

Moderator

4 Feb 2026 13:34



LO9 (Planning and organising work)

LO9 (Planning and organising work)

Grading criteria shown for each TMA and the EMA. The text in blue is an addition from me. Please note how the grading criteria changes from TMA to TMA, and from TMA03 to the EMA.

	TMA01	TMA02	TMA03	EMA
LO9 (Planning and organising work) Plan and organise your project work appropriately, and keep systematic records of plans, progress and outcomes. <i>7 marks</i>	Has selected an appropriate lifecycle for the project having considered all the alternatives in light of the characteristics of their specific project. Has given an appropriate schedule of future work that reflects their lifecycle and the characteristics of their chosen project.	Has a plan and makes an accurate assessment of progress in relation to the original plan. Understands what has gone well and what has not gone to plan and, more importantly, why.	Has a clear plan and makes an accurate assessment of progress in relation to the original plan. Understands what has gone well and what has not gone to plan [and, more importantly, why].	Has clearly planned (choice of lifecycle & time plan) and accurately managed progress in relation to the original plan. Understands what has gone well and what has not gone to plan [and, more importantly, why.]

LO9 Plan and organise

your project work appropriately, and keep systematic records of plans, progress and outcomes.

EMA grading criteria ...

Has clearly planned (choice of lifecycle & time plan) and accurately managed progress in relation to the original plan. Understands what has gone well and what has not gone to plan.

Project planning and management on TM470 projects tends to be weak. All students should revisit previous OU modules in which project planning and management was covered if appropriate, as well as use the TM470 module resources ("Planning and Organising a Project") and do any additional research if required. Also, you can contact your tutor if you need any project management or planning guidance.

The first phase of a project is always project initiation, and the last phase is always project close down. Of course, people or organisations may use different titles or terms for these phases.

Has clearly planned (choice of lifecycle & time plan) and accurately managed progress in relation to the original plan. You can only demonstrate this if you have actually kept a copy of the original plan rather than having over-written it when making updates. You should keep copies of all significant versions of project plans so you can discuss progress, updates, revisions etc. And you might want to include one or more plans in appendices. LO9 does say "... keep systematic records of plans, progress and outcomes.", so this should not come as a surprise.

You should also be reporting progress to stakeholders and your tutor. I ask my own students to produce a monthly report. A sample of a report should be included in an appendix, and maybe a list of dates on which reports were sent out.

Understands what has gone well and what has not gone to plan. To adequately cover this you need to talk about key elements (not all!) that have gone well and some that have not gone well. In order to do this effectively, you should reference and talk about specific named tasks, activities, milestones, deliverables, dates etc in your plan.

To do effective project management, you need to identify all deliverables, create a work breakdown structure (WBS), estimate effort/duration for all the deliverables from the tasks in the WBS. Prince 2 also includes a product breakdown structure (PBS), a product basically being a fancy word for a deliverable.

Most plans I see lack granularity and have lots of missing elements. For example, tasks for acquiring knowledge and or skills to enable you to complete the project should be in your plan somewhere. A lot of the more academic content is often missing. For example, carrying out the literature search/review and the LSEPI/EDI analysis. The TM470 TMA's can have an impact upon your technical project timelines, as can doing TMAs for other modules. These should appear in your project plan somewhere, if nothing else than to remind you of other commitments/dependencies. Holidays, and other periods when you are not available for working on the project should also appear on the plan.

Occasionally, I see some plans that don't provide dates but use week numbers instead. And sometimes they also don't provide a start date and end date for the project. You should use actual dates so the reviewer of your plan can see straight away the timelines in your plan.

We are not trying to turn you into project managers, but we do expect you to demonstrate that you have basic knowledge of project management and put it into practise on your project. Having said this, project management is a sought-after skill and is highly remunerated in the IT industry. So, it might be beneficial for you to use your project as a good opportunity to put into practise and demonstrate project management skills.

Updated versions of Plans for the EMA (my reply to a query from a student)

You need to keep the initial plan and create an updated copy. You don't want to include complete versions of all your updated plans in appendices. This will make the EMA too long and difficult to navigate. Have the initial plan in an appendix and the final one in a main chapter. If the final plan is far too big/long, then you might only show main phases, major deliverables, milestones, etc and leave out finer details, lower-level tasks, etc and include the full final plan in an appendix. Also, have a list of dates when you created updated versions along with a description of key/major changes in an appendix.

For intermediate plans (between initial and final), you could take a screenshot if you wish to explain some major/key change or update.

If you had a major issue and had to significantly replan your project with significant changes, then you might want to include that full version in an appendix too.



Kawal Banga

Moderator

18 Feb 2026 10:42



Back up your project

Hi all,

You must back up all work on a regular basis. You could store it in the cloud, on an external hard drive or a USB stick, email it to yourself or a combination. If you are using a company laptop, don't assume it is backed up, or don't assume that even if it is backed up, they will be able to retrieve the data. Try retrieving some old files to see if the backups are working. Most companies won't back up your c: drive, so don't store any data/documents on the c: drive of a company laptop.

The TMAs/EMAs state "Throughout your project you should make regular backups, perhaps to a cloud service or an external hard drive. This is your responsibility, and the University cannot make allowances for loss of data resulting from hardware failure."

What's prompted this post? A project student emailing me last night, telling me that his company laptop has crashed on the day he is supposed to submit his TMA02, assignment and he's been storing his documents on the C Drive! So he can't continue working on his TMA on the temporary laptop they've given him. Let's hope that IT support rescue his data from the laptop. Just imagine if this happens when you are about to submit your EMA! You might end up failing the module.

Kawal
